

STAGE 1 — KNOWLEDGE & DATA HARVESTING

Knowledge-Driven Target Discovery

PubMed mining (endogenous mediators)
+ CTD (migraine targets)

Intersection: 4 shared targets
CALCA, CALCB, TAC1, TNF

A. E-MTAB-13397 — Whole blood RNA-seq

N=104 (52 subj); Interictal vs Control
Repeated measures

B. PRJEB40032 — PBMC RNA-seq

N=42 (32 subj); Interictal/Ictal vs Basal

C. GSE198274 — Serum exosomal miRNA

N=30 (10 Ctrl, 10 Pre, 10 Post)
Paired cases

STAGE 2 — ANALYSIS & MULTI-OMICS INTEGRATION

Subject-aware ML
feature selection

LASSO / RF / SVM-RFE
Internal AUC = 0.531
External AUC = 0.483
Inv. AUC = 0.517

miRNA-mRNA integration
& pathway profiling

miRNA baseline: 38
(FDR<0.05)
Post-tx delta: 0
(FDR<0.05)

Two-sample
Mendelian randomization

Genetic triangulation
Nominal signals
FDR>0.05
(no robust drivers)

STAGE 3 — RESULTS & INTERPRETATION

32-gene migraine signature

Intersection:
ML-selected $\cap$
Acupuncture-linked

Molecular subtyping (k=3)

General: C1=7, C2=15, C3=8
Acupuncture: C1=9, C2=16, C3=5

Conclusion (conservative)

Findings are consistent with state-dependent molecular heterogeneity and downstream response pathways rather than primary inherited drivers.